

Database Schema Description

Overview

The Employee Management System consists of four main entities:

- Department
- Employee
- Project
- EmployeeProject

The system supports department management, employee management, project management, and employee-project assignments.

Department

Stores information about company departments.

Column	Type	Description
id	Long	Primary Key
name	String	Department name
location	String	Department location
budget	Double	Department budget

Relationships

- One Department can have many Employees.
 - One Department can have many Projects.
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Employee

Stores employee information.

Column	Type	Description
id	Long	Primary Key
name	String	Employee name
email	String	Employee email
phone	String	Employee phone number
hireDate	LocalDate	Employee hiring date

Column	Type	Description
salary	Double	Employee salary
department_id	Long	Foreign Key referencing Department

Relationships

- Each Employee belongs to one Department.
- An Employee can be assigned to multiple Projects through EmployeeProject.

Project

Stores project information.

Column	Type	Description
id	Long	Primary Key
name	String	Project name
description	String	Project description
startDate	LocalDate	Project start date
endDate	LocalDate	Project end date
department_id	Long	Foreign Key referencing Department

Relationships

- Each Project belongs to one Department.
- A Project can have multiple Employees through EmployeeProject.

EmployeeProject

Represents the many-to-many relationship between Employees and Projects.

Column	Type	Description
id	Long	Primary Key
employee_id	Long	Foreign Key referencing Employee
project_id	Long	Foreign Key referencing Project
role	Enum	Employee role within the project

Relationships

- Each record links one Employee to one Project.

- An Employee can participate in multiple Projects.
 - A Project can contain multiple Employees.
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Relationship Summary

- Department (1) → (N) Employee
- Department (1) → (N) Project
- Employee (M) ↔ (N) Project through EmployeeProject